

hCRI Companion – User Manual

Covers app version 1.20 · Last updated October 9, 2026 · [Source on GitHub ↗](#)

01 Overview

hCRI Companion is an independent Android app that connects to a Hopoocolor HPCS-series spectrometer (or a Torch Bearer, via the ESP32 bridge) over Bluetooth Low Energy, reads the full spectrum off the sensor, computes the same colorimetric values hCRI.io itself computes, and uploads a reading to your hCRI.io account in one tap – no cable, no CSV export, no re-typing numbers off the meter's own tiny screen.

Every CCT, Duv, Ra, Rf, Rg, and R1–R15 value shown in the app is computed with the exact same algorithm hCRI.io's report page uses, so a reading never disagrees with itself between your phone and your account.

Supported meters

HPCS-310

HPCS-310P

HPCS-330

HPCS-330P

HPCS330Pro

Torch Bearer (via ESP32 bridge)

The app detects which model you're connected to from its advertised Bluetooth name and reads its data with that model's own field layout automatically – nothing to configure. The full, up-to-date list (including which models are independently verified against a real captured reading) is always on the app's own Settings → About tab.

The Torch Bearer needs the ESP32 bridge

The Torch Bearer has no Bluetooth of its own. A LilyGO T-Display S3 board running the open-source hCRI bridge firmware connects to it by USB and talks to the app over Bluetooth. The [hcri-esp32-bridge repository](#) lists the hardware you need and explains how to install the firmware. Hopoocolor HPCS meters connect directly and need no bridge. See [Using a Torch Bearer](#).

What's new in 1.20

- In-app TM-30 report: after any reading (including a test reading) a **TM-30** button opens the full IES TM-30-18 Color Rendition Report on your phone – scroll through it, or tap **Share PDF** at the bottom to send it as a PDF. It is calculated in the app, so it works offline and for readings that were never uploaded. The **TM-30** button in History now opens this report too.
- Fixed the drag and drop in Settings → Main Screen Measurements (rows no longer jump around), and **Reset to Default** now moves the rows back.

What's new in 1.19

- More reliable update checks: the app re-checks Google Play each time you return to it, and Settings → Update shows what Play reported (the newest and installed version codes), which is also written to the Logs tab.

What's new in 1.18

- Layout fix for the Galaxy Z Fold 8 Ultra: the bottom tab labels (Home, History, Settings) and the action-bar labels above them are no longer cut off at the bottom of the screen.

What's new in 1.17

- Updates now come from Google Play. The app checks Play for a newer version when it opens, and the banner (and Settings → Update) has a button that opens the app's Play Store page, where you install the update. The old in-app update screen is gone.

What's new in 1.16

- Flicker readings now upload to hCRI.io: with a reading (attached to its report) or on their own, with a title and notes.
- A redesigned Flicker tab: `New sample`, `Add to reading` / `Redo for reading`, one shared `Stop`, a `?` help button and the flicker settings gear in the top-right of the chart.
- The bottom Flicker button only opens the Flicker tab; Flicker is hidden while Live is running.
- History and Data actions are now icons with captions (cloud = upload to hCRI.io, tray = share, eye = view, trash = delete); readings that include flicker show a small waveform icon.
- On the T-Display-S3 Pro, the bridge firmware shows battery percentage and can power a Torch Bearer from its own battery.

What's new in 1.15

- Torch Bearer support, with an optional spectral correction (Settings → Torch Bearer correction).
- Continuous (Live) readings and Flicker on meters that support them.
- Device scanning, connecting and switching between meters refined a lot.
- History: Share selected readings, a restyled top bar, and a title editor whose Save works on the first tap.
- Shared CSV files now end with a short summary of the computed values.

- Flicker readings can now be uploaded to hCRI.io (see 1.16 above).
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02 Requirements

- Android 8.0 (Oreo) or newer
- Bluetooth Low Energy (BLE) support
- A Hopoocolor HPCS-series meter, powered on and in range – or a Torch Bearer connected to the [hCRI ESP32 bridge](#) (see [Using a Torch Bearer](#))
- An hCRI.io account, if you want to upload readings (not required just to take and view them)

03 Installing the app

hCRI Companion isn't on the Play Store – it's a direct install, the same way most spectrometer and BLE utilities in this space are distributed.

1 Download the APK

From the [Companion download page](#), tap Download APK. Android downloads the file like any other.

2 Allow installs from this source

Opening the downloaded file for the first time, Android will ask to confirm installing from an unknown source – this is expected for any app distributed outside the Play Store. Allow it for your browser or Files app.

3 Install and open

Finish the install prompt, then open hCRI Companion from your app drawer.

Updating later

If you installed from Google Play (including the closed test), the app checks Play for a newer version each time it opens. When one is available, a banner appears and Settings → Update shows `Open in Play Store`; the update itself is installed from the Play Store page. If you installed the APK directly, there's no automatic check: re-download and install the APK the same way to get a newer build – it installs over the existing app and keeps your saved readings, account, and settings.

04 Connecting your hCRI.io account

You only need to do this once. From the `Settings` tab, there are two ways to get a token into the app.

Option A: Generate a token in the app

- 1 Tap the  button next to Save**
Opens the Generate API Token dialog.
- 2 Enter your hCRI.io username or email, and your password**
Used once to sign in – your password is never stored on the device and never sent anywhere else.
- 3 Review the token name**
Defaults to "hCRI Companion Data Upload Token" – this is the name you'll see later if you list or revoke tokens from Profile → API on the website. Change it if you'd like something more specific.
- 4 Tap Generate & Save**
The app signs in, creates a brand-new token under that name, and saves it – same as entering one by hand (Option B, step 3 below).

Option B: Paste in an existing token

- 1 Enter your hCRI.io username**
The same username you sign in with at hcri.io.
- 2 Enter your hCRI.io API token**
Find or generate this from Profile → API on hcri.io. The field is masked while typing.
- 3 Tap Save**
The token is stored in your device's secure keychain (Android Keystore) – not in plain text, and not readable by any other app.

Either way, once saved, the fields switch to a locked, read-only view with a `Forget Credentials` button – a deliberate extra step before replacing a saved key, so a stray keystroke can't silently overwrite a working token.

Why generate instead of paste?

It skips a trip to the website – sign in once in the app and you're done. Each token you generate (in the app or on the website) is independent and can be revoked on its own from Profile → API without affecting any other device or token.

05 Pairing with your meter

The app looks for your meter automatically when it opens. You can also go to the **Main** tab and tap **Connect to Meter**. The app scans for a few seconds and connects automatically if exactly one supported meter is in range.

If Bluetooth is turned off on your phone, a message tells you that Bluetooth is required to take readings, with an **Open Settings** button that jumps to your phone's Bluetooth settings and an **OK** button that simply closes it. This message appears when the app opens and when you tap **Connect to Meter** – not every time you switch back to the app.

If more than one matching meter is found, a picker appears so you choose which one – look for its advertised name and signal strength. A small **≡ Switch meter** icon next to the status row lets you reopen that same list later without a fresh scan. Device switching has been refined a lot in 1.15: the list stays current while you look at it, a meter that is momentarily busy is retried automatically, and switching to another meter no longer needs a reset.

No meter handy? See [Try it without a meter](#).

Meter won't connect?

Tap **Meter won't connect? Reset connection**, underneath the Connect button. This clears a stale Bluetooth connection left over from a previous session – the most common reason a meter seems stuck, and something that used to require power-cycling the meter itself.

06 Using a Torch Bearer

The Torch Bearer spectrometer reaches the app through the hCRI ESP32 bridge. You need:

- a LilyGO T-Display S3 board (the standard version, not the Pro, with 16 MB flash)
- a USB-C OTG cable or adapter with a USB-A socket for the Torch Bearer plus a USB-C socket for power
- a power bank to run the bridge in the field
- the Torch Bearer spectrometer, and this app on your phone

Full hardware notes, ready-built firmware downloads and the flashing steps are in the [hcri-esp32-bridge repository](#). Once the bridge is flashed and powered, it shows up in the app as `Torch Bearer` and connects like any other meter.

Good to know

- The bridge allows one Bluetooth connection at a time and stops advertising while connected – close other Bluetooth tools (such as nRF Connect) if the app can't see it.
- Its exposure adapts automatically. Hold the sensor about 25 cm from the lamp, ideally behind a diffuser; very close to a bright source it can saturate.
- **Torch Bearer correction** (Settings, on by default) adjusts the spectrum so CCT, Duv, Ra and R9 line up with the HPCS. In our checks on five lamps it cut the CCT error from about 208 K to 36 K. Turn it off for the raw spectrum; every shared CSV records whether it was applied.
- Lux and PAR are not calibrated yet – treat them as approximate (within about 15 percent of an HPCS).
- The Torch Bearer takes single readings; Live and Flicker are not available on it.

07 Live readings & Flicker

On meters that support them, the action bar shows a **Live** button and a **Flicker** button. **Live** keeps measuring and updates the spectrum and values continuously until you tap **Stop**. Buttons for modes your meter doesn't support simply don't appear. While Live is running the Flicker button and the Flicker chart page are hidden; they return when you stop Live.

The Flicker tab

The **Flicker** button on the action bar only takes you to the Flicker tab (the fourth chart page on Main); it never starts anything. The tab shows the flicker frequency (Hz), percent flicker, flicker index, cycle time and duty, the captured waveform (or a risk chart, with the Wave / Risk switch) and a risk tip. The **?** in a circle at the top right explains the options below, and the gear next to it sets the meter's flicker range and sample rate.

There are two kinds of flicker sample, each started with its own button under the chart:

- **New sample** measures flicker on its own, continuously. It *clears the current reading* from Main – a reading that was saved stays in History, a live reading you had not saved is gone. When you tap **Stop** you can **Share** the sample or **Upload flicker** it to hCRI.io (see below). A New sample is not added to History.
- **Add to reading** (shown as **Redo for reading** when the reading already has a sample) measures flicker for the reading you just took. The last frame when you tap **Stop** becomes that reading's flicker sample: it is saved with the reading, shown in History, and uploaded together with the reading. It therefore has no Share or Upload of its own.

While a sample runs, its button pulses and says *Sampling...*, the other button dims, and one red **Stop** sits between them. A counter shows how many updates have arrived and the elapsed time.

When each option is available

- After a reading that has not been uploaded: both **Add to reading** / **Redo for reading** and **New sample**.
- After the reading has been uploaded: only **New sample**, because an uploaded report can't change.
- No reading on screen (or after a New sample): only **New sample**, with Share and Upload flicker once a sample exists.
- Imported or test readings: only **New sample**.

- Flicker that came with a reading never shows Share or Upload flicker, and the readings you open from History show their flicker without any controls.
- A note under the buttons reminds you that New sample clears the current reading, and whether it is saved to History.

Capture flicker with each reading

Settings → *Capture flicker with each reading* (off by default) takes a one-shot flicker snapshot automatically after every reading. It adds a few seconds per reading. Even when it is off you can add a sample to a reading yourself with `Add to reading`, before you upload it.

Uploading flicker to hCRI.io

- **With a reading:** when you upload a reading that has a flicker sample, the report goes up first and the flicker is then attached to it automatically. It appears in the flicker section at the bottom of the report on hCRI.io.
- **On its own:** after a New sample, `Upload flicker` (cloud icon) opens a small window for a title (up to two lines) and optional notes. Both are remembered and pre-filled next time. The flicker then appears under Explore → Flicker on hCRI.io, where it can be viewed, compared with other flicker readings, and attached to a report.
- The flicker upload is best-effort: if hCRI.io can't take it, the report upload is not affected, and the Logs tab says what happened.

Flicker readings are also included at the end of shared CSV files.

08 Taking a reading

Once connected, tap `Take Reading`. The meter takes its exposure, and a short haptic buzz confirms the moment it lands (a different pattern if it fails) – handy when you're not looking directly at the screen. The spectrum, CCT, Duv, Ra, and the rest of your chosen measurements appear immediately after.

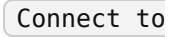
Every completed reading is saved automatically to [History](#) – there's nothing extra to do to keep it.

09 The Main tab

Main is where you connect, take readings, and review the latest one. Its result card always shows the same layout – a grid of measurements, the three swipeable charts, and the action button directly underneath them – whether or not a reading has happened yet (before your first reading, everything just reads zero/blank, so nothing jumps around once real numbers appear).

A small  button beside the Main / Data / Logs tabs opens a short overview of what the app does. Tapping the  tab at the bottom of the screen always brings you back to Main, even from Data or Logs. The phone's bottom navigation bar is hidden to give the app more room; swipe up from the bottom edge to bring it back for a moment. While the keyboard is open the bar stays visible, so Back and Home remain reachable.

Try it without a meter

While no meter is connected, a  button appears under . It loads the spectrum from a random public report on hCRI.io, so you can explore the measurements and charts before you own a meter. Tap  under the button for a short explanation.

A test reading is sample data, not yours: it is not saved to History, and Upload (on both Main and Data) is disabled while one is showing. Its title is read-only, and a note under the buttons names the report it came from. Connect a meter and take a real reading to replace it. The button disappears as soon as a meter is connected.

Measurement grid

Shows CCT, Ra (CRI), Duv, Lux, Rf, R9, and Rg by default. Which measurements appear here, and in what order, is fully customizable from [Settings](#) – including any individual R1–R15 sample and the raw CIE x/y coordinates.

Spectrum / Chrom / R-Values

Swipe (or tap the dots) between three views of the same reading:

- **Spectrum** – the full wavelength-by-wavelength power distribution, colored by wavelength, plus a collapsible list of every raw nm/value pair.
- **Chrom** – the CIE 1931 chromaticity diagram, with your reading's x/y point plotted against the real spectral locus and the Planckian (blackbody) curve.
- **R-Values** – the full CRI R1–R15 breakdown as a color-coded bar chart, the same chart hCRI.io's own report page uses.

The TM-30 report

Once a reading is on screen – including a [test reading](#) – a **TM-30** button appears at the right of the bottom bar (and on every History row). It opens the full IES TM-30-18 Color Rendition Report: Rf and Rg, CCT, Duv, Ra, R9 and chromaticity, the color vector graphic, the spectrum against its reference illuminant, local chroma shift, hue shift and fidelity for the 16 hue bins, and the fidelity of all 99 color samples. Scroll the page (pinch to zoom). The **Share PDF** button docked at the bottom creates a PDF of the same report and opens your share sheet. The report is calculated on your phone with the same math hCRI.io uses, so it needs no upload and no connection.

Uploading

Tap **Upload to hCRI.io** to send the current reading. A small green checkmark appears next to the button once it succeeds – there's no popup to dismiss on a routine upload. If an upload fails, an alert explains what happened; check the [Logs tab](#) for the server's own response if you need more detail.

10 The Data tab

A closer look at the current reading, plus the controls for naming and sending it:

- **Upload Title** – type a custom title, or leave it blank to use the auto-generated username + timestamp + device label.
- **Upload Preview** – the exact wavelength/value payload about to be sent, so you can eyeball the range and point count before it goes anywhere.
- **R1–R15, Chromaticity**, and **Device Info** – full numeric detail, including the meter's own firmware version and integration time.
- **Upload to hCRI.io** and **Share CSV** – the same upload button as Main (with the same inline checkmark), plus exporting this one reading as a CSV file through Android's share sheet.

11 The History tab

Every reading you've ever taken, grouped by day, newest first. From here you can:

- **View** a past reading's full measurement grid, spectrum, chromaticity, and R-values exactly as they looked when it was taken.
- **Rename** a reading – renaming before upload sends it to hCRI.io under that new title.
- **Re-upload** a reading at any time, including ones already sent. Row actions are icons: cloud (upload to hCRI.io), tray (share CSV), eye (view) and trash (delete). A small waveform icon next to the CCT/Ra marks a reading that includes a flicker sample, which goes up with it.
- **TM-30, Copy, Report** – every reading has a  TM-30 button that opens its TM-30 report in the app (see [The TM-30 report](#)), whether or not it was ever uploaded. For a reading that has been uploaded,  puts its hCRI.io link on the clipboard and  Report opens the report in your browser. For a private report, the first tap creates its share link, which anyone holding the link can open.
- **Select** multiple readings and upload them to hCRI.io in one batch, instead of opening and sending each individually.
- **Share** a single reading, the readings you have selected, or your entire history, as CSV. Shared files end with a short summary of the computed values.
- **Delete** a reading you no longer want – this only removes it from your phone; a copy already uploaded to hCRI.io stays there until you delete it from your account separately.

12 The Logs tab

A running debug log of connect/disconnect/measure/upload activity, each line timestamped in your phone's own local time. Useful for troubleshooting, or to attach to a bug report.

- **Share Debug Log** – sends the current log through Android's share sheet (email, messaging, wherever you'd like to send it).
- **Clear** – empties the log.

Turning on **Verbose logging** in Settings adds the low-level Bluetooth hex traffic and the full raw result body to this log – off by default, since the standard log already covers ordinary connection and measurement status.

13 Settings

hCRI.io account

Your username and API token – see [Connecting your hCRI.io account](#) above.

Appearance

Light, Dark, or System (the default) – follows your phone's own light/dark setting unless overridden here.

Stay connected in background

On by default. Keeps the meter connected when you switch away from the app or lock the screen, so coming back doesn't need a fresh reconnect. Turn this off to disconnect the meter whenever the app isn't in the foreground instead – saves the meter's own battery, at the cost of reconnecting (a few seconds) every time you come back. See [Background & reconnect behavior](#) for the full picture.

Keep screen awake while connected

Prevents the screen from locking during a session with the meter connected. Uses more battery – off by default.

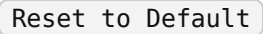
Verbose logging

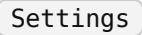
Adds the full wavelength-by-wavelength and Bluetooth hex detail to the debug log – see [The Logs tab](#).

Torch Bearer correction

On by default. Applies the spectral correction to Torch Bearer readings so CCT, Duv, Ra and R9 agree with the HPCS. Turn it off to see the raw spectrum. Does not affect HPCS meters.

Main Screen Measurements

Check which measurements show on the Main tab's result card, and hold the  handle to drag a row and reorder it.  restores CCT, Ra, Duv, Lux, Rf, R9, Rg in that order.

Settings has two tabs at the top:  and , so About is one tap away instead of at the bottom of a long scroll.

About

The current list of supported meter models (and which are independently verified), a [Send Feedback](#) button that opens a pre-addressed email, and a link back to this Companion page.

14 Background & reconnect behavior

By default, switching away from hCRI Companion – the app switcher, locking your screen, a phone call – leaves the meter connected ([Stay connected in background](#), on by default in Settings). Coming back to the app picks up right where you left off, with no reconnect delay.

If you'd rather the app disconnect the meter whenever it isn't the one on screen (trading a short reconnect delay for a little extra meter battery life), turn that setting off. Either way, coming back to the foreground with nothing connected always tries to reconnect automatically – you shouldn't need to tap Connect yourself just because the meter dropped on its own while you were away.

Sharing a CSV or the debug log briefly hands control to Android's own share sheet, which looks like backgrounding the app – this is recognized and never triggers a disconnect either way, since you're still right there using the app.

15 Troubleshooting

Meter won't connect, or seems stuck

Tap `Meter won't connect? Reset connection` on the Main tab (shown while disconnected). This clears a leftover Bluetooth connection from a previous session – by far the most common cause, and something that used to require power-cycling the meter.

App crashed, or won't open

A crash lands on a "Try Again" screen rather than kicking you back to your home screen. If the app won't stay open at all, force-close it from Android's recent-apps view and reopen it – this app recovers from a mid-session crash cleanly.

A reading looks wrong

Double-check the correct meter model connected (shown next to the status dot on Main) – each model has its own field layout, and the wrong one can occasionally still connect if its advertised name happens to be ambiguous. If a value still looks off, check Settings → About for whether that model is marked independently verified.

Still stuck?

From the Logs tab, tap `Share Debug Log` and send it along with a description of what happened to fizzgig@hcRI.io, or use `Send Feedback` on the About tab of Settings.

16 Privacy

hCRI Companion doesn't collect, store, or transmit anything about you on its own – no analytics, no tracking, no third-party server in the loop. A reading lives only on your device until you tap Upload, which sends it straight to your own hCRI.io account over an encrypted connection and nowhere else. If you use `Generate Token` in Settings, your password is sent once, over that same encrypted connection, to sign in and create the token – it's never stored on the device. Full details are in the [Privacy Policy](#) on the Companion page.

17 Feedback & support

Questions, bug reports, or feature requests are always welcome at fizzgig@hcri.io – the same address the app's own [Send Feedback](#) button (Settings → About) opens for you.

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